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### Modelling species presence-only data with random forests

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The random forest (RF) algorithm is an ensemble of classification or regression trees and is widely used, including for species distribution modelling (SDM). Many researchers use implementations of RF in the R programming language with default parameters to analyse species presence-only data together with 'background' samples. However, there is good evidence that RF with default parameters does not perform well for such 'presence-background' modelling. This is often attributed to the disparity between the number of presence and background samples, also known as 'class imbalance', and several solutions have been proposed. Here, we first set the context: the background sample should be large enough to represent all environments in the region. We then aim to understand the drivers of poor performance of RF when models are fitted to presence-only species data alongside background samples. We show that 'class overlap' (where both classes occur in the same environment) is an important driver of poor performance, alongside class imbalance. Class overlap can even degrade performance for presence-absence data. We explain, test and evaluate suggested solutions. Using simulated and real presence-background data, we compare performance of default RF with other weighting and sampling approaches. Our results demonstrate clear evidence of improvement in the performance of RFs when techniques that explicitly manage imbalance are used. We show that these either limit or enforce tree depth. Without compromising the environmental representativeness of the sampled background, we identify approaches to fitting RF that ameliorate the effects of imbalance and overlap and allow excellent predictive performance. Understanding the problems of RF in presence-background modelling allows new insights into how best to fit models, and should guide future efforts to best deal with such data.

Keywords: class imbalance, class overlap, down-sampling, ecological niche model, presence-background, recursive partitioning

#### Introduction

Species distribution modelling (SDM) is central to ecological and conservation studies. SDM combines species occurrence data and environmental variables to predict species spatial distributions (Franklin 2010). Many modelling methods are used for fitting SDMs (Guisan et al. 2017). While earlier examples of SDM mostly focussed on regression methods (Guisan et al. 2002), attention has turned towards the application



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of machine learning (ML) methods (Elith et al. 2006). This is mainly due to their flexible fitted functions, automatic variable selection methods, ability to handle different types of data, and frequent reports of powerful predictive performance (Elith et al. 2006, Olden et al. 2008, Hastie et al. 2009). Amongst the plethora of ML methods available for modelling and mapping species distributions, random forests (RF; Breiman 2001) has become very popular (Cutler et al. 2007, Zhang et al. 2019) because it tends to predict well with minimal parameter tuning (Freeman et al. 2016, Probst et al. 2019b). RF is also viewed as useful because it inherently includes interactions in the fitted model (Elith 2019), and estimates of variable importance are readily available (but see Strobl et al. 2007). Of course, there are also drawbacks associated with high flexibility (Merow et al. 2014), and since RF offers little control over the shape of the fitted function and interaction level, the potential for such models to be overfitted needs to be taken seriously and evaluated robustly (Roberts et al. 2017).

Whilst RF are often reported to perform well, there are cases where RF has predicted species distributions poorly, often linked to use of presence-only species data fitted as binary data by using background samples as the second class (and hereafter called presence-background data). Background locations sample the landscape. Intuitively, background samples allow SDM methods to compare environmental conditions favoured by the species to those available throughout the landscape under consideration. The word ‘pseudo-absence’ is sometimes used interchangeably with ‘background’ (Barbet-Massin et al. 2012) or sometimes means sampling from all places where the species has not been recorded (Araújo et al. 2019). Here we focus on background samples as a sample of the landscape, because the statistical meaning of that sampling is clear (Warton and Shepherd 2010, Fithian et al. 2015, Renner et al. 2015).

Several authors have demonstrated poor predictive performance of RF compared to other methods when using presence-background data in SDMs (Johnson et al. 2012, Shabani et al. 2016). This weak performance has been attributed to the large number of background samples (Barbet-Massin et al. 2012, Liu et al. 2013). This is not a general problem related to presence-background models, since many methods such as Maxent and Boosted Regression Trees (BRT) have been shown to perform strongly on these types of data (Elith et al. 2006). This is a specific problem for RFs. Poor performance has also been reported for RF when modelling presence-absence data containing a large number of absences (Evans and Cushman 2009, Freeman et al. 2012, Robinson et al. 2018). This behaviour of RF is well-known within ML, where it is similarly often linked to the sensitivity of these models to imbalanced classes, i.e. where one class (majority class, here background points) substantially outnumbers the other class(es) (minority class, here presence points) (Chen et al. 2004, He and Garcia 2009, Khalilia et al. 2011).

Although several SDM studies have reported issues with RFs (Barbet-Massin et al. 2012, Liu et al. 2013, Robinson et al. 2018), and have suggested solutions, none focus on explaining

the problem and the merits and limitations of the proposed solutions. Here we address this gap, by explaining how RFs work and how common features of SDM data impact their performance. We focus on presence-background data, but also show when to expect problems with presence-absence datasets. We use the ML literature to shed light on the underlying issues. We test how readily available solutions suggested in both the SDM and the machine learning literature work with simulated data, and with a real dataset.

## Random forests: an overview

A Random Forest is an ensemble of classification or regression trees (CART). CARTs are based on recursive partitioning methods (Box 1). The recursive partitioning repeatedly splits the data into potentially high-dimensional rectangular partitions of the covariate (predictor) space, choosing those for which the response data are relatively homogenous (Strobl et al. 2009). Individual trees can have high variance, i.e. a small change in the training data changes the fitted tree, leading to poor generalisation. An ensemble of many trees, however, generalises better (Hastie et al. 2009). RF fits many individual trees, usually several hundred to thousands, and combines their predictions (Strobl et al. 2009, Elith 2019). Depending on the response variable, RF uses classification trees (CTs) or regression trees (RTs) for qualitative and quantitative responses, respectively. Each tree is fitted on a bootstrap sample of the training data – a random sample, taken with replacement, with a total sample size equal to the number of records in the training data. On average, each bootstrap sample contains 63.2% unique records (Efron and Tibshirani 1994), called in-bag samples. Samples not selected are called out-of-bag samples and are used to estimate the model’s error.

Unlike other closely related ensembling methods such as bagging (Elith 2019), on each split RF only evaluates a random subset of predictors to identify the best predictor, instead of searching over all predictors (Box 1). This produces decorrelated trees, making overfitting less likely (Breiman 2001, Strobl et al. 2009). The number of predictors evaluated at each split is represented by the parameter *mtry* in many RF packages. RF by default grows ‘deep’ trees, i.e. unpruned trees with many splits (Strobl et al. 2009, Elith 2019). This can lead to few observations in terminal nodes. Minimum observation in terminal nodes and depth can be controlled by model parameters, though control options depend on the implementation.

## Prediction with RFs for binary data

Here we focus on how the predicted values are estimated in RFs, since binary data can be treated in two ways. If treated as classes (hereafter named ‘classification-RF’), for any one tree the outcome at each terminal node is predicted as the most frequent class across the observations in that node, also known as majority vote (Breiman 2001, Strobl et al. 2009). The classification-RF prediction class for any observation is

### Box 1. Tree growing and splitting criteria

Decision trees are grown by splitting the response data on predictor variables. On a chosen predictor each split point is selected to separate the response into two nodes, aiming to achieve the highest purity/similarity within each node, and the highest impurity/dissimilarity between nodes. On each split, all predictors are tested and the predictor causing the highest impurity gain is chosen. Each level of splits is a tree depth. The impurity is calculated with measures such as the Gini index for classification trees (CTs). For binary data and CTs, the Gini index is calculated as  $2p(1 - p)$ , where  $p$  is the proportion of one of the classes (e.g. in species data, presence and absence/background are the classes) (Hastie et al. 2009). The splitting criterion for regression trees (RTs) is usually calculated using a sum of squares instead of the Gini index. Taking the average of the response  $y$  as the predicted value  $\bar{y}$  in each node, sum of squares is calculated as  $\sum (y_i - \bar{y})^2$  over all  $i$  observations (Hastie et al. 2009).

Breiman (2001, 2002) proposed the Gini index as the default splitting criterion for CTs (as single trees and within RFs). The Gini splitting criterion is now known to have weaknesses when applied to datasets with high-class imbalance (Drummond and Holte 2000, Cieslak and Chawla 2008). Alternative criteria have been developed that are relatively insensitive to class imbalance such as Hellinger distance (Cieslak and Chawla 2008). The Hellinger distance was first implemented in decision trees by Cieslak and Chawla (2008), known as Hellinger distance decision tree (HDDT). This method has been shown to be a robust criterion and insensitive to class imbalance and is claimed to be superior to the traditional splitting criteria such as Gini when used on imbalanced data (Cieslak et al. 2012).

then calculated again by majority voting, now across all fitted trees. For any prediction, class probabilities can also be calculated (Liaw and Wiener 2002) as the proportion of the predicted positive class ('1') of all CTs in the final classification-RF model (Breiman 2002). Binary data can also be modelled with RTs (Malley et al. 2012, Zhang et al. 2019), hereafter 'regression-RF'. This treats the response as continuous, and the mean over all observations in a terminal node is the predicted node outcome, which is then averaged over all trees in the final regression-RF (Hastie et al. 2009). Given the different splitting criteria (Box 1) and approaches to calculating predictions across trees, predicted probabilities for binary responses will not necessarily be identical between RTs and CTs (Malley et al. 2012). Both prediction approaches are implemented in the R package `randomForest` (Liaw and Wiener 2002). Whilst most modellers use classification-RF for binary data, Malley et al. (2012) found that probabilities estimated by regression-RF are more accurate and consistent than those from classification-RF.

The R package `ranger` (Wright and Ziegler 2017) is a fast implementation of RFs. For modelling binary data, they use the ideas of Malley et al. (2012), naming the resulting regression-RF as a 'probability forest'. Instead of using the traditional sum of squares split criteria (Box 1), for probability forests they use the Gini index or Hellinger distance for splitting. Predictions are estimated as described above for regression-RF (Wright and Ziegler 2017).

## Understanding class imbalance

The predictive performance of RF can be substantially affected by class imbalance. There has been considerable interest in reasons and solutions for these impacts in the ML literature, for both RF and other algorithms (Chen et al. 2004, He and

Garcia 2009, Ali et al. 2015, Krawczyk 2016). Unequal representation of the classes is known to cause problems (He and Garcia 2009, Khalilia et al. 2011). Other issues associated with imbalanced data include a small sample size of the minority class, small disjuncts within the minority class and class overlap (Japkowicz 2003, Denil and Trappenberg 2010, Ali et al. 2015). We do not expand on the small samples and small disjuncts here because we expect they are less relevant to the SDM presence-background problem than imbalance and overlap.

Class overlap occurs in covariate space, when two classes co-occur over the same multi-covariate ranges. It is also referred to as class complexity or class separability (Japkowicz and Stephen 2002, Prati et al. 2004). Class overlap is often the most serious problem in imbalanced data (Prati et al. 2004, Denil and Trappenberg 2010, Ali et al. 2015). When classes overlap in some or all of the covariate space it is very hard to determine discriminative rules to separate the classes (Ali et al. 2015). The splitting procedure of trees tries to separate the response into two groups that may not be completely separable with the available predictors. Since an RF model by default can fit very deep trees, the algorithm can continue to attempt to separate the response to reduce impurity and may go to the level of individual observation, thus overfitting to the training data.

### A demonstration with presence-absence data

Class overlap will by definition occur in presence-background data, because the background points are meant to sample the full environmental space in the region, including that in which the species occurs (Renner et al. 2015). However, it can also occur in presence-absence data if presences and absences are recorded in similar environments. This can occur, for instance, in species that are sparse (low abundance)

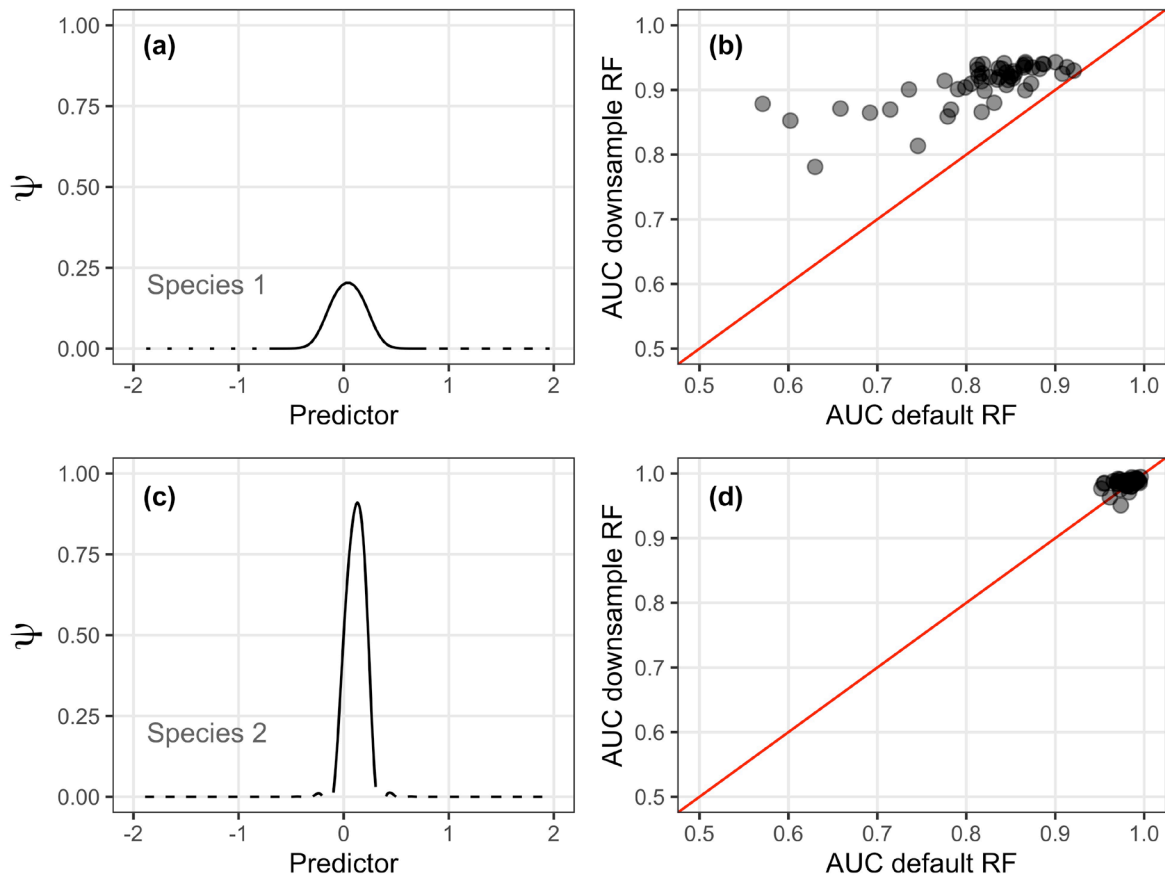


Figure 1. Two species distributions simulated for assessing the impact of class overlap on RF. The left plots (a, c) show the probability of species occurrence ( $\psi$ ) as a function of a single predictor. We assume the predictor is equally represented across the landscape. (b) and (d) show the predictive performance of RF with default settings versus RF using a method for dealing with class imbalance (down-sampling, see section ‘Suggested solutions’). See ‘Data availability’ section for code.

over a large range of their prime habitat (Rabinowitz 1981). Meynard and Kaplan (2012) describe it as a gradual response to the environment, with many intermediate values and very few extreme (very low or very high) probabilities. Since presence–absence data are simplest to conceptualise, we start with a presence–absence simulation.

We simulated two presence–absence datasets for two virtual species (Fig. 1, see ‘Data availability’ section for code). Both datasets are severely imbalanced (species prevalence below 0.06, i.e. less than 6% of presence class) and both species occur in only a small range of the predictor variable. Species 1 (Fig. 1a) has a gradual response to the environment and a fairly low occurrence probability even in the most suitable environments, so the species is both present and absent in its prime habitat, demonstrating class overlap. In contrast, species 2 (Fig. 1c) has a steep response to environment and a high prevalence within the narrower range of suitable environments, leading to minimal class overlap (very few absences recorded within prime habitat). We generated 50 datasets of 300 records (presence and absence) from each species. We model using the `randomForest` package with default settings (classification-RF, fitting 500 trees to all records) and also with an implementation of RF that handles class

imbalance by down-sampling the majority class (explained in following sections). Predictive performance is estimated on 1500 independent records for each species.

Here we evaluate with the area under the receiver operating characteristic curve (AUC) because, as explained later, this is a relevant measure when we move to presence-background data. For species 1, neither approach to fitting RFs achieves an average AUC very close to 1 (as expected for such distribution characteristics; Meynard and Kaplan 2012), but the performance of the down-sampling approach (AUC ranging from 0.78 to 0.94) is always better than that of the default RF (AUC from 0.57 to 0.92, Fig. 1b). Addressing class imbalance/overlap improves prediction (see next section). In contrast, for species 2 with high probability of presence within prime habitat (Fig. 2c; little class overlap yet still very low prevalence across all predictor values and hence high class imbalance), both approaches to fitting an RF model give AUCs close to 1 (Fig. 1d). The contrast between results for species 1 and 2 shows that it is not the imbalance of classes itself that impacts predictive performance, but class overlap. This simple example illustrates that when classes can be clearly separated by the available environmental conditions, commonly used default methods for fitting RF models perform

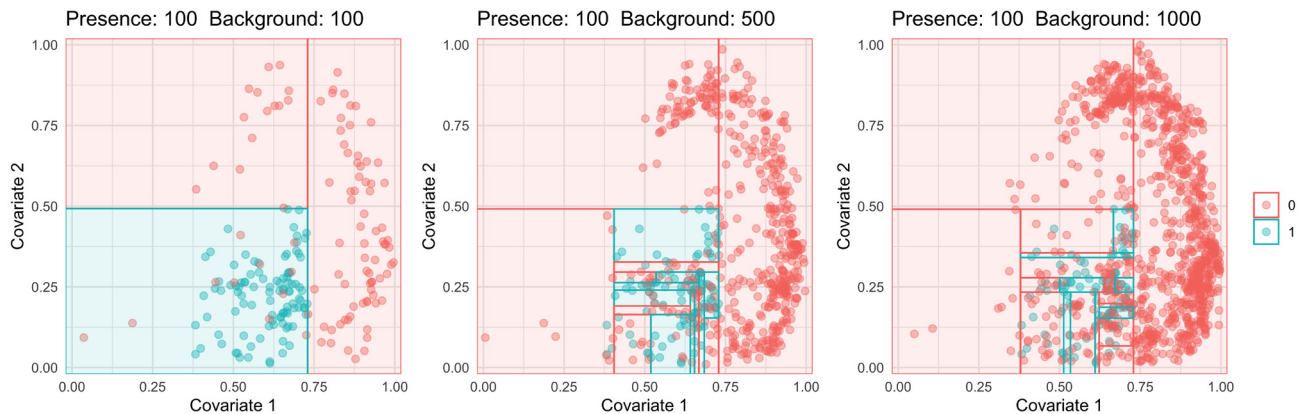


Figure 2. Visualisation of recursive partitioning with different number of background points. The 0s (red) and 1s (blue) show background and presence samples, respectively. The blue sections are the terminal nodes predicted as presence and red are predicted as background. The lines in the plots show the split points. This is a tree fitted by `rpart` package (Therneau et al. 2019) with maximum allowed depth of 25 and complexity parameter of 0, i.e. no early stopping imposed.

well, regardless of class imbalance. Similar results are reported by Japkowicz and Stephen (2002) that decision trees (and two other ML algorithms) are less sensitive to class imbalance when classes are separable.

### The problem for RF and presence-background data

In presence-background data the two classes are presence and background, and class overlap is expected because background samples aim to sample all environments, including those preferred by a species. With background samples obtained randomly and independently of species locations, there is potential for exact spatial and therefore environmental overlap (i.e. background points can be at the same location as presence points). Furthermore, background samples should characterise environmental conditions in the region of interest, so potentially tens of thousands of samples are needed, depending on the complexity of environments (Franklin 2010, Renner et al. 2015). So, when modelling presence-background data, it is common to experience both severe class imbalance and class overlap. The solution is not to sample fewer background points (risking not properly representing environmental conditions) or to ensure that presence and background samples do not co-occur, because all that does is to move away from the statistically correct way to create these samples. A different solution is needed.

To understand how tree-based models fit presence-background samples, we simulate presence and random background samples across two covariates (Fig. 2; see ‘Data availability’ section for code). We fit CTs to the data, with the algorithm allowed to generate a deep tree if warranted by the data (similar to RF). With a 1:1 ratio of presence to background (left panel), despite some class overlap, the CT stops at a simple two-split model because no improvement can be made by further splits. As more background points are added, more overlap and more imbalance is inevitable, so the algorithm continues splitting to find small patches where

presences are still dominant. Note here that, with presence-background data, class imbalance and class overlap both necessarily come into play. It has been shown for support vector machines, which also focus on separating classes, that imbalance and overlap tend to interact, and result in overly complex models (Denil and Trappenberg 2010). That is likely occurring here. Since we know from theory that many background samples are needed to properly represent the background environmental conditions, what is the solution?

### Suggested solutions

Here we focus on solutions relevant to RF fitted to presence-background data. We include approaches suggested by both SDM and ML researchers. Whilst there are several proposed solutions in the ML literature, we only explore methods readily available in R packages, and also those that suit the problem of modelling spatially explicit presence-background data (Discussion). Most solutions focus on tweaking the model or resampling the data. Code for implementing these solutions in both `randomForest` and `ranger` packages are provided in the Supporting information.

### Applying weights

This approach deals with class imbalance and arose in the ML literature. Weights are applied to each class, using the notion of cost-sensitive learning. Typically the misclassification cost for the minority class is increased, by providing a higher weight (Chen et al. 2004). Weights are estimated in different ways across software implementation of RFs. We use the popular `randomForest` package (Table 1) (Liaw and Wiener 2002), in which the weighting is applied to the Gini index, impacting selection of splits and also weighting across terminal nodes in each tree (Chen et al. 2004). In the `randomForest` package weights are only implemented for classification-RF.

Table 1. Implemented RF model parameters and descriptions. The training data for all models were the same except for the equal-sample model, which was run 10 times with a reduced background sample.

| RF model                                            | Parameters  | Value                           | Description                                                                                                                | R package    |
|-----------------------------------------------------|-------------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------|
| Default (classification RF)                         | ntree       | 1000                            | number of trees                                                                                                            | randomForest |
| Regression (RF)                                     | ntree       | 1000                            | number of trees                                                                                                            | randomForest |
| Weighted (classification RF)                        | ntree       | 1000                            | number of trees                                                                                                            | randomForest |
|                                                     | classwt     | Inverse weights                 | class weights: inverse weights were calculated as 1/proportion (class).                                                    |              |
| Equal-sample (classification RF)                    | ntree       | 1000                            | number of trees<br>model is run 10 times with equal number of presence and background samples and predictions are averaged | randomForest |
| Down-sample (classification RF)                     | ntree       | 1000                            | number of trees                                                                                                            | randomForest |
|                                                     | sampsize    | same as the number of presences | size of the sample to draw from each class for each tree                                                                   |              |
|                                                     | replace     | TRUE                            | take bootstrap sample with replacement                                                                                     |              |
| Shallow (regression RF or probability forest)       | num.trees   | 2000                            | number of trees                                                                                                            | ranger       |
|                                                     | splitrule   | Hellinger                       | Hellinger distance splitting criterion                                                                                     |              |
|                                                     | max.depth   | 2                               | maximal tree depth (= maximum 4 terminal nodes)                                                                            |              |
|                                                     | probability | TRUE                            | use ranger probability forest                                                                                              |              |
|                                                     | replace     | TRUE                            | take bootstrap sample with replacement                                                                                     |              |
| Shallow-tuned (regression RF or probability forest) | num.trees   | 2000                            | number of trees                                                                                                            | ranger       |
|                                                     | splitrule   | hellinger                       | hellinger distance splitting criterion                                                                                     |              |
|                                                     | max.depth   | 1–4                             | maximal tree depth                                                                                                         |              |
|                                                     | probability | TRUE                            | use ranger probability forest                                                                                              |              |
|                                                     | replace     | TRUE                            | take bootstrap sample with replacement                                                                                     |              |

## Equal-sampling

This idea arose in the SDM literature, in the context of presence-background modelling. The idea of equal-sampling (Barbet-Massin et al. 2012) is to fit  $n$  different classification-RF models (where  $n$  varies with data but is often 10) on  $n$  different subsamples of the data set, each with all presences and a random background sample of size equal to the number of presences. The final prediction is averaged over the  $n$  models. This is based on ‘repeated random subsampling’, a ML technique for dealing with imbalanced classes (Japkowicz and Stephen 2002, Khalilia et al. 2011). It works by reducing the number of background samples (the majority class) and thus reducing the opportunity for class overlap. Since the number of background samples is matched to the number of presence samples in each repeat, the total number of background samples used in the final model will be, at maximum,  $n$  times the number of presences. For small presence samples, this has the undesirable consequence that relatively few background samples are taken unless  $n$  is increased and thus computation time is also increased. We used  $n = 10$  (Table 1).

## Down-sampling

Down-sampling is similar to the equal-sampling approach, but it differs in that it subsamples at the level of the individual tree. It was first suggested in the ML literature (Chen et al. 2004) then used for SDM (Evans and Cushman 2009, Freeman et al. 2012, Robinson et al. 2018, Valavi et al. 2021). It uses the same number of background samples as presence

points in each CT in a classification-RF model by sampling with replacement (bootstrapping) from the full background set. Since forests typically have hundreds of trees, overall this approach uses many more background samples than the equal-sampling technique (above) as each individual tree uses potentially a different set of background samples. As for equal-sampling, down-sampling works by balancing the training data (Chen et al. 2004). Down-sampling is often referred to as balanced RF (Chen et al. 2004, Robinson et al. 2018).

## Using regression-RF instead of classification-RF

The approaches described above use classification-RF reflecting the fact ML practitioners often model binary data as a classification task (Krawczyk 2016). However, regression is also relevant to binary data. Malley et al. (2012) provide evidence that class probabilities are better predicted via regression-RFs rather than classification-RFs. Zhang et al. (2019) recommend using regression-RF for presence-background modelling. We use no weights on samples, and all presence and background points are used.

## Using shallow probability trees (with Hellinger distance splitting)

Our visualisation in Fig. 2 shows that trees will tend to try to separate overlapping classes, and with large background samples, this implies trees will become deeper and deeper – i.e. the model will become increasingly complex. Here we propose an approach that deals directly with this issue, focussing on the issue of class overlap. Our idea is to enforce shallower

trees. In addition, we choose a splitting criterion known to be robust to imbalanced data. Within the options of commonly available R packages, this suggests using Hellinger distances in preference to the Gini index (Drummond and Holte 2000, Cieslak and Chawla 2008, Cieslak et al. 2012). One way to bring these ideas together is to fit an RF in the `ranger` package (Wright and Ziegler 2017), since this enables probability forests and control of tree depth. In this approach, no subsampling is required, and all background samples are used in each probability tree of the forest.

## Tests with real and simulated data

### Material and methods

#### Species data for modelling

To assess the predictive performance of these approaches, we used a global dataset (hereafter ‘the NCEAS dataset’) of 225 species from six regions of the world, fully described and made available by the NCEAS data group (Elith et al. 2020). This dataset has presence-only species data for model fitting and presence-absence survey data for model evaluation. It comprises birds and plants of the Australian Wet Tropics (AWT); birds of Ontario, Canada (CAN); birds, plants, mammals and reptiles of north-east New South Wales, Australia (NSW); plants of New Zealand (NZ); plants from five countries of South America (SA); and plants of Switzerland (SWI). The number of presence-only records ranges from 5 to 5822 per species and 7 to 12 environmental predictors were available. The evaluation datasets range from 102 (AWT) to 19 120 (NZ) survey sites (Elith et al. 2020). Since the presence-only data are known to be biased (Phillips et al. 2009), we aim to deal with that bias in our modelling. Rather than using a random background sample, we aim for a background sample with similar biases to that in the presence-only records. Following Kujala et al. (2015), we generated 50 000 background samples from a bias layer constructed using kernel density estimation on target-group samples (i.e. all species in each biological group) defined by Phillips et al. (2009) (see full description and code in the Supporting information – this also includes an alternative implementation with the standard target-group background).

In addition, we generated 100 virtual species with the `virtualspecies` package ver. 1.5.1 (Leroy et al. 2016). Real environmental variables from Australia were used to simulate these species. We generated a species suitability map by applying a Gaussian function (with random mean and standard deviation for each species) on the first four principal component axes of 21 environmental covariates (see ‘Data availability’ section for code). The final suitability was a multiplicative function of responses to the four predictors. Then rasters of presence-absence data for modelling and evaluation were randomly drawn following a binomial distribution with suitability values at each pixel as success rates. Depending on the prevalence of each simulated species, from 20 to 182,

presence points were randomly drawn as presence-only data. In addition, 50 000 random background samples of the landscape were taken and used for all species. For evaluating predictions, 5000 random presence-absence samples were drawn for each species. These were specified to exclude all training set presence sites and their direct neighbours (cells). The environmental covariates in both datasets were tested for collinearity and for those pairs with a Pearson correlation coefficient higher than 0.8, one variable was removed.

#### RF implementations

We implemented all the strategies described above (‘Suggested solutions’) alongside a `randomForest` classification-RF. Default parameters were used except for the number of trees, which we changed to 1000 to be consistent with the other solutions, and following recommendations in the ML literature to use a large number of trees (Probst et al. 2019b). Models were implemented in the R programming language ([www.r-project.org](http://www.r-project.org)) using `randomForest` ver. 4.6-14 (Liaw and Wiener 2002) and `ranger` ver. 0.12.1 (Wright and Ziegler 2017). The parameters and implementations are described in Table 1, and code of all strategies for both packages is provided in the Supporting information.

We did not tune all the parameters of the models as RFs are known to work reasonably well with default parameters (Freeman et al. 2016, Probst et al. 2019a). For the shallow probability trees, we used the package `ranger` and chose ‘maximum depth’ of 2 (i.e. maximum 4 terminal nodes; column 1, Table 1). Since the choice of 2 is subjective, to assess the sensitivity of this parameter, we fitted another shallow model and tuned the maximum depth (allowing depths of 1–4) using a 5-fold cross-validation on the presence-background training data. As the aim to keep the trees shallow is to avoid overfitting, we did not try values higher than 4 ( $2^4 = 16$  possible terminal nodes). In comparison, the maximum depth (and thus number of terminal nodes) for trees in the `randomForest` implementations can be very high, and with default settings can lead to only 1 observation per terminal node in classification-RF and 5 for regression-RF. We recorded number of terminal nodes in all models for the NCEAS datasets (details and code in the Supporting information).

#### Model evaluation and comparison

Two threshold-independent evaluation metrics were used to assess predictive performance of the models: Area under the ROC curve (AUC) and Pearson correlation (COR). AUC is a well-known metric in SDM. It measures discrimination – i.e. the ability of the model to discriminate presences from absences (Pearce and Ferrier 2000). COR values were calculated 1) between the predicted relative likelihood of occurrence and the presence-absence records of the evaluation data, in the NCEAS dataset; and 2) between the predicted likelihood of occurrence and the simulated probability of occurrence, for the virtual species. Note that we are not

including metrics that directly measure calibration, because presence-background methods cannot be properly calibrated by nature of the data inputs (Guillera-Arroita et al. 2015).

To determine whether the modelling approaches had statistically different performance, we used non-parametric hypothesis tests for comparing models over multiple datasets (Demšar 2006, García and Herrera 2008). The Friedman's Aligned Rank test (García and Herrera 2008) was calculated for AUC (and COR) using the Bergmann–Hommel correction for p-values (Bergmann and Hommel 1988).

## Results

### Performance on real datasets

The performance of the models on the real datasets (NCEAS) is shown in Fig. 3. Considering AUC of 0.7 or higher as useful discrimination (Pearce and Ferrier 2000), the default RF had poor performance in all regions except SWI. It was the poorest performer overall, followed by weighted and regression-RF. The down-sample, equal-sample and both shallow RF models performed noticeably better for both AUC and COR, for all regions (Fig. 3). The Friedman's aligned rank test was highly significant, with adjusted p-values showing: 1) the difference in down and equal samples is not significant, and they are slightly and significantly better than both shallow models; 2) shallow and shallow-tuned are not significantly different; 3) all these four models are significantly better than the default, weighted and regression-RF (Supporting information). The values of tree depth chosen by the tuning favoured depth of 4 (190 species were fitted with depth 4 and only 8 with depth 2). The average number of terminal nodes for equal and down-sampled methods was always substantially lower than that of default classification-RF and regression-RF (Supporting information). These trends in performance were generally consistent

across regions, though the differences between methods in some regions (e.g. SA) were starker than in others (e.g. SWI, where differences were minor).

### Performance on simulated datasets

Figure 4 shows the performance of the models on 100 simulated species as evaluated with AUC and COR. The default model had poor performance on both evaluation metrics, with a median AUC ~0.62 and median COR less than 0.2. The weighted classification-RF had a very similar performance and showed no improvement. The two (equal and down) sampling approaches and the shallow RFs clearly improved performance (Fig. 4), achieving median AUC higher than 0.85 and median COR higher than 0.75. Both shallow approaches produced slightly better results than other models. Across all species the selected tree depth in the shallow-tuned model varied from 1 to 4, with similar numbers of each (25, 21, 25 and 29 times for depth 1–4, respectively). However, the tuned version of shallow RF showed no evident performance boost over the shallow with constant max-depth of 2. The regression-RF showed much better discrimination than default and weighted classification-RF, but very similar COR. This is in line with the result found by Zhang et al. (2019).

The spatial prediction of implemented methods on an example virtual species is presented in Fig. 5, with all scaled to vary from 0 to 1 (see code in the Supporting information). The default, weighted and regression RFs predicted tightly around the presence sites (very small dots in their map), implying an extreme case of overfitting. The average number of terminal nodes of trees (refer to as t. node in Fig. 5; Supporting information) shows that these three models have deep trees (149, 186 and 306, respectively), presumably overfitting the data. On the other hand, down-sample, equal-sample and shallow closely follow the spatial pattern of the virtual species, and have less deep trees (28, 22 and 3, respectively).

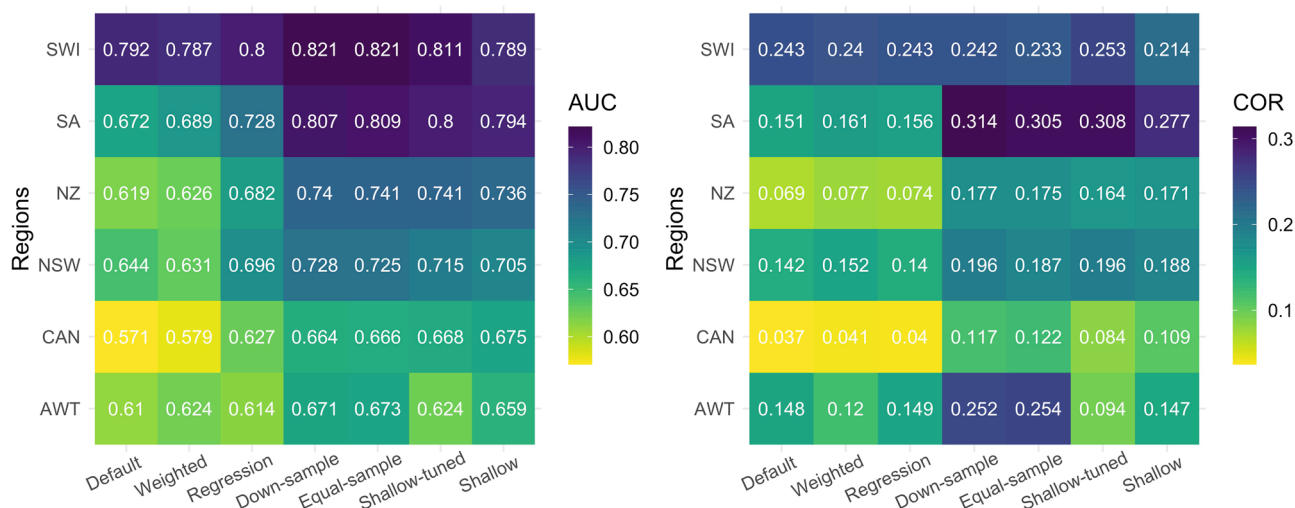


Figure 3. Performance of different RF models on different regions of NCEAS data. The values in cells are AUC and COR for the left and right plots respectively. Colour varies with values, yellow (low) to dark navy (high).

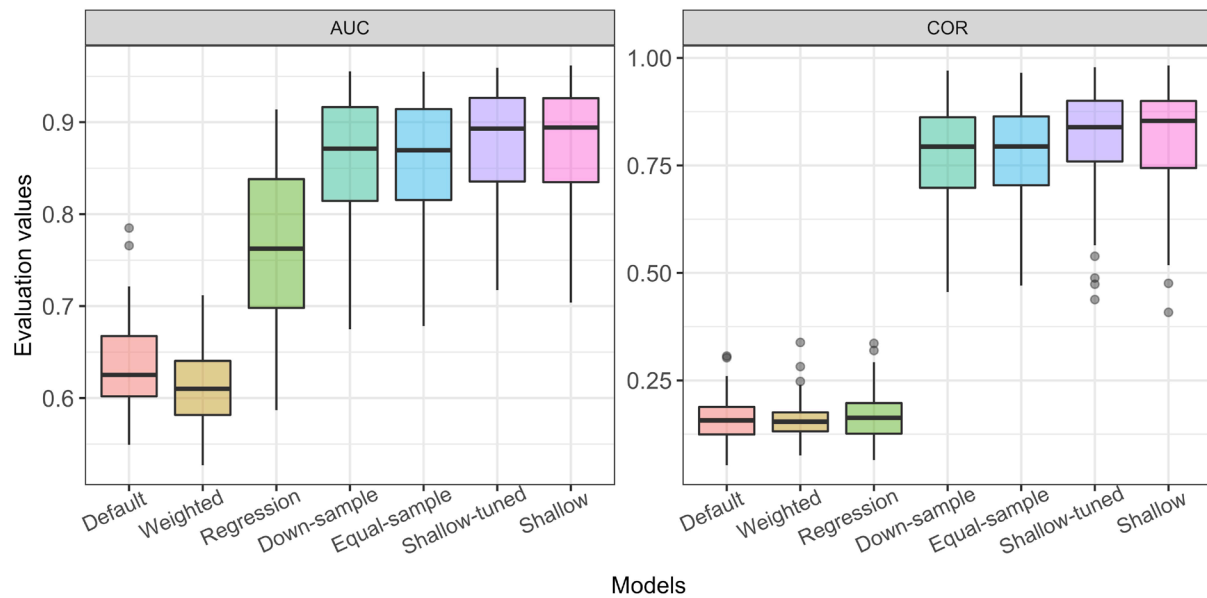


Figure 4. Performance of different RF models on 100 virtual species. The left plot is the AUC and the right one is COR values.

The Friedman's aligned rank test was highly significant, with p-values showing shallow and shallow-tuned are not significantly different, but are significantly better than down-sample and equal-sample. These four models are not significantly different from one another in COR, but are significantly better than other methods in both metrics (Supporting information).

## Discussion

We have explained the problems facing RF models fitted to presence-only data with background samples. We remind readers that these problems are specific to RF, and not widely seen even in related methods such as BRTs (Valavi et al. 2021). Several solutions (equal-sampling, down-sampling and shallow

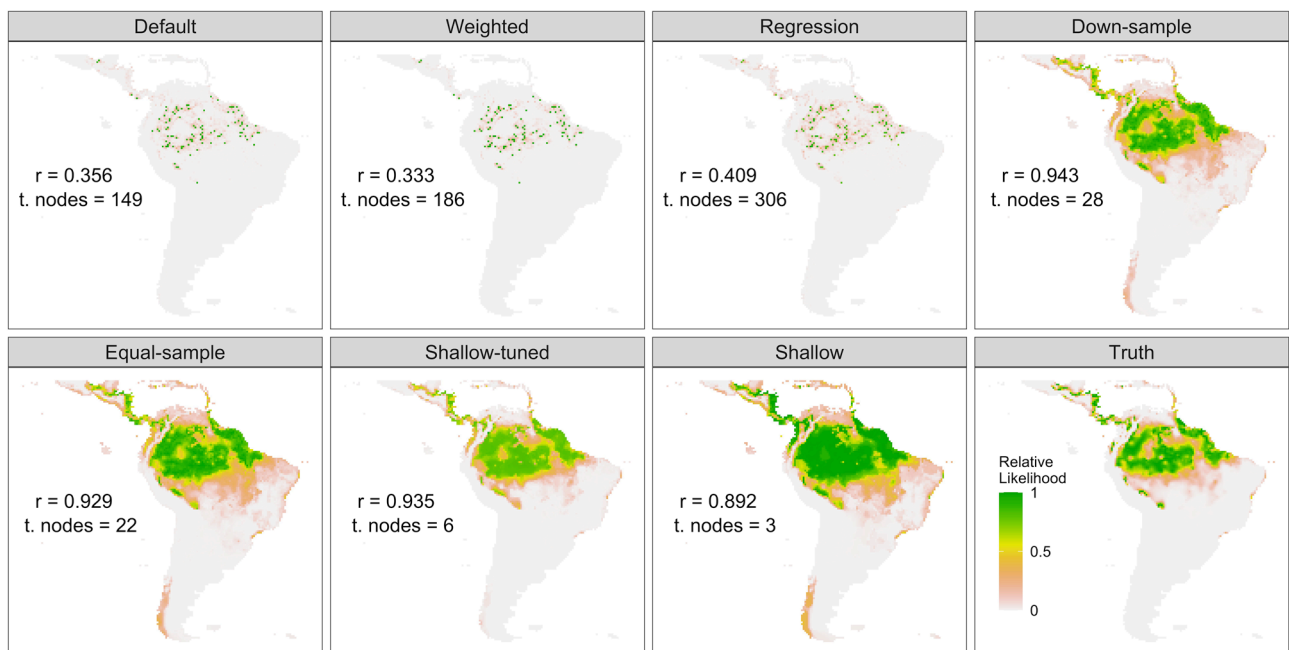


Figure 5. The spatial prediction of different RF implementations on a virtual species. The ' $r$ ' shows the correlation between the virtual species (Truth) and models predictions. The 't. nodes' shows the number of terminal nodes of each model averaged over all trees. Code and more description about virtual species are provided in the Supporting information. For ease of comparison, the likelihood of all the maps were linearly rescaled between 0 and 1, so if there are a few high predictions and the rest is low, the map will be pale (i.e. overfitting). Models here are implemented based on packages and parameters presented in Table 1.

RF) improved performance significantly. Here we focus on those solutions, interpreting their approach and performance in the light of the presence-background modelling problem.

### Effectiveness of sampling methods for class imbalance

Down-sampling and equal-sampling provided similar results, and both substantially improved predictions in both real and simulated datasets, and reduced overfitting by fitting less deep trees than default settings (Supporting information). Both approaches deal directly with the amount of class imbalance and overlap by subsampling the majority class (background), but they do it differently. Down-sampling focuses on data used in each tree (Chen et al. 2004), whereas equal-sampling provides subsampled sets at the level of the forest, repeated (Barbet-Massin et al. 2012). This means that down-sampling is faster for prediction to rasters and covers more of the full set of background samples. Based on theory, methods that use more background samples are preferable, up to the point that all environmental space in the region is well represented (Franklin 2010, Renner et al. 2015). Indeed, for all presence-background models there are good statistical reasons for providing enough background samples to cover the environmental space (usually tens of thousands), sampling them irrespective of presence locations (Warton and Shepherd 2010, Renner et al. 2015). This implies that – even though there is no noticeable performance benefit here of down-sampling over equal-sampling, one would expect that in some situations the additional environmental information in larger samples would be beneficial. The research of Barbet-Massin et al. (2012), which suggested equal-sampling, is widely cited and has influenced ideas of how to fit RF with presence-background data. Our results here suggest a benefit of moving from that to the faster and easier implementation of down-sampling. RF down-sampled has been successfully used in several ecological studies to address the challenge of class imbalance (Evans and Cushman 2009, Robinson et al. 2018, Shaeri Karimi et al. 2019).

Whilst resampling is commonly used for handling imbalanced data in ML (Cieslak et al. 2012), and – where used for SDMs – has shown benefits (Evans and Cushman 2009, Barbet-Massin et al. 2012, Robinson et al. 2018, Valavi et al. 2021), improvement is not universal. For instance, Freeman et al. (2012) evaluated the effectiveness of down-sampling on presence-absence data of tree species distributions. They did not find any improvement with the down-sampling technique over default RF. We wonder whether there was low-class overlap in their data, and note that it would be useful to have measures to quantify and report this (Mthembu and Marwala 2008, Lorena et al. 2019).

Other methods of resampling have been proposed in the ML literature, taking the opposite approach and up-sampling the minority class (Kuhn and Johnson 2013). Since with presence-background datasets the true data are the presences and the background samples are simply a technique for fitting

the models, we chose to leave the true data untouched and only resample background points. In addition, presence-only records often suffer from spatial sampling bias, thus duplicating presence points or synthetically generated new presence samples (Robinson et al. 2018) might exacerbate this problem. Hence, we do not see a strong argument for testing these other methods for presence-background modelling.

### Another approach of handling class imbalance in RFs

In this study we tested a new approach, based on our understanding of the dual impacts of class imbalance and class overlap (Denil and Trappenberg 2010), and our desire to use all background samples. In our shallow RF approach, we used probability forest (a regression-RF) with shallow trees and Hellinger distance splitting criterion without any subsampling i.e. all background samples were used in model fitting. The shallow method predicted well (Fig. 3, 4). However, forcing the model to fit shallow trees deviates from one the main characteristics of RF model which is using unpruned trees (Breiman 2001, Strobl et al. 2009). We suggest these shallow RF models need further investigation. The relative roles of the splitting criterion and tree depth in improving performance are unclear, though some testing for our supplementary code examples showed that implementing shallow trees with a Gini index in the `randomForest` package did not perform well. This implies that tree depth and a splitting criterion insensitive to imbalance are both important. It would also be interesting to test methods for restricting the depth of trees in regression-RF, since they performed better than default methods in AUC, without any adjustments like resampling, weights or limiting tree depth.

### Final remarks

When applied on presence-background data the default implementation of RF model shows relatively poor predictive performance. RFs focus on separating the response into two groups, and allow deep trees to do this. Hence, they have an intrinsic issue with data that have overlap between classes, such as presence-background data. We show RF can be used effectively to fit SDMs by adjusting the implementation to accommodate the realities of the data. The fastest, most straightforward and well-tested implementation amongst those we tested is down-sampling ('balanced RF') as introduced by Chen et al. (2004) and used for SDM by Evans and Cushman (2009) and Freeman et al. (2012), and SDM users could confidently use that for presence-background modelling.

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### Author contributions

**Roozbeh Valavi:** Conceptualization (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Validation

(lead); Visualization (lead); Writing – original draft (lead); Writing – review and editing (equal). **Jane Elith:** Conceptualization (equal); Formal analysis (supporting); Methodology (equal); Supervision (equal); Writing – original draft (equal); Writing – review and editing (equal). **Jose Lahoz-Monfort:** Conceptualization (equal); Formal analysis (supporting); Methodology (equal); Supervision (equal); Writing – original draft (equal); Writing – review and editing (equal). **Gurutzeta Guillera-Arroita:** Conceptualization (equal); Formal analysis (supporting); Methodology (equal); Supervision (lead); Writing – original draft (equal); Writing – review and editing (equal).

## Transparent Peer Review

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## Data availability statement

All environmental grid data and simulation codes are available via Open Science Framework (OSF): <<https://osf.io/dkmw6>>. The Supporting information is available at: <<https://rvalavi.github.io/SDMwithRFs>>).

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